

Write a program in **C++/Java** that performs a **DFS** on the following graph and outputs edges with labels: **tree edge, back edge, forward edge, and cross edge**. Submit your program and a screenshot of the outputs in a single **Word/PDF** file.

Your program should accept the following file as an input. You have to write code to process this input. You should define and use **adjacency list representation** of graph.

The input in the file is in this format:

- First line is number of nodes.
- Rest of lines are in this format: `node number (node degree) : adjacency list`